Example 1.3.6 : *For a binary memory source with two symbols x_1 and x_2, show that entropy $H(X)$ is maximum when both x_1 and x_2 are equiprobable. What are the lower and upper bounds on $H(X)$?*

Solution : (i) To prove that x_1 and x_2 are equiprobable :

Let the probability of x_1 be, $P(x_1) = p$.

$\therefore \quad P(x_2) = 1-p$

Following steps are required to solve the problem :

(i) Determine entropy $H(X)$.

(ii) Differentiate $H(X)$ with respect to p, i.e. $\frac{d}{dp} H(X)$.

(iii) Solve $\frac{d}{dp} H(X) = 0$ for p.

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(ii) To obtain entropy of source i.e. $H(X)$:

Entropy is given as,

$$H(X) = \sum_{i=1}^{M} P(x_i) \log_2 \frac{1}{P(x_i)}$$

$$= P(x_1) \log_2 \frac{1}{P(x_1)} + P(x_2) \log_2 \frac{1}{P(x_2)}$$

$$= p \log_2 \frac{1}{p} + (1-p) \log_2 \frac{1}{1-p}$$

We know that $\log_2 \frac{1}{x} = -\log_2 x$. Hence above equation becomes,

$$H(X) = -p \log_2 p - (1-p) \log_2 (1-p)$$

(iii) To determine $\frac{d}{dp} H(X)$:

$$\frac{d}{dp} H(X) = \frac{d}{dp}[-p\log_2 p-(1-p)\log_2(1-p)]$$

$$= -\frac{d}{dp}[p\log_2 p]-\frac{d}{dp}[(1-p)\log_2(1-p)]$$

$$= -\left[\log_2 p+p\frac{d}{dp}\log_2 p\right]-\left[\log_2(1-p)(-1)+(1-p)\frac{d}{dp}\log_2(1-p)\right]$$

We know that $\log_2 x=\frac{\log_e x}{\log_e 2}=\frac{\log_{10} x}{\log_{10} 2}$. Then above equation becomes,

$$\frac{d}{dp} H(X) = -\left[\log_2 p+p\frac{d}{dp}\frac{\log_e p}{\log_e 2}\right]-\left[-\log_2(1-p)+(1-p)\frac{d}{dp}\frac{\log_e(1-p)}{\log_e 2}\right]$$

We know that $\frac{d}{dx}\log_e x=\frac{1}{x}$. Hence above equation will be,

$$\frac{d}{dp} H(X) = -\left[\log_2 p+p\cdot\frac{1}{p}\cdot\frac{1}{\log_e 2}\right]-\left[-\log_2(1-p)+(1-p)\cdot\frac{1}{(1-p)}(-1)\cdot\frac{1}{\log_e 2}\right]$$

$$= -\left[\log_2 p+\frac{1}{\log_e 2}\right]-\left[-\log_2(1-p)-\frac{1}{\log_e 2}\right]$$

$$= -\log_2 p-\frac{1}{\log_e 2}+\log_2(1-p)+\frac{1}{\log_e 2}$$

$$= -\log_2 p+\log_2(1-p)$$

(iv) To determine value of p for $\frac{d}{dp} H(X)=0$:

When the derivative of $H(X)=0$, it has maximum value. Hence by $\frac{d}{dp} H(X)=0$, we obtain value of p.

At this value of p, $H(X)$ is maximum. i.e.,

$$0 = -\log_2 p+\log_2(1-p)$$

$$\therefore \quad \log_2 p = \log_2(1-p)$$

$$p = 1-p$$

or $$p = \frac{1}{2}$$

Thus $p(x_1) = p=\frac{1}{2}$

and $p(x_2) = 1-p=\frac{1}{2}$

(v) To obtain upper and lower bounds on $H(X)$:

We know that $H(X)$ is non-negative. Hence lower bound on $H(X)$ will be '0'. In example 1.3.4, it is proved that upper bound on $H(X)$ is $\log_2 M$. i.e.,

$$0 \leq H(X) \leq \log_2 M$$

Above equation gives lower and upper bound on $H(X)$. Here 'M' is the number of messages emitted by the source.

Example 1.3.7 : *For a discrete memoryless source there are three symbols with probabilities $p_1 = \alpha$ and $p_2 = p_3$. Determine the entropy of the source and sketch its variation for different values of α.*

Solution : The three probabilities are,

$$p_1 = \alpha \quad \text{and} \quad p_2 = p_3$$

We know that, $p_1 + p_2 + p_3 = 1$

$$\therefore \quad \alpha + p_2 + p_2 = 1 \quad \text{since} \quad p_2 = p_3$$

$$\therefore \quad p_2 = \frac{1-\alpha}{2}$$

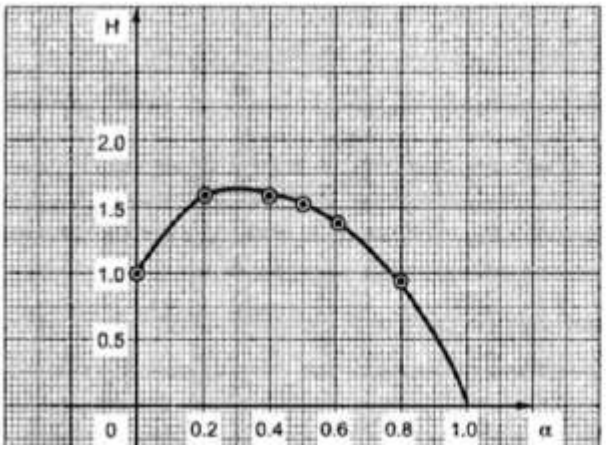
Hence $$p_2 = p_3 = \frac{1-\alpha}{2}$$

$$H = \sum_{k=1}^{3} p_k \log_2 \frac{1}{p_k}$$

$$= \alpha \log_2 \frac{1}{\alpha} + \left(\frac{1-\alpha}{2}\right) \log_2 \left(\frac{2}{1-\alpha}\right) + \left(\frac{1-\alpha}{2}\right) \log_2 \left(\frac{2}{1-\alpha}\right)$$

$$= \alpha \log_2 \frac{1}{\alpha} + (1-\alpha) \log_2 \left(\frac{2}{1-\alpha}\right)$$

The sketch of H with respect to α is given in Fig. 1.3.2



α	H
0	1
0.2	1.52
0.3	1.58
0.4	1.57
0.5	1.5
0.6	1.37
0.8	0.92
1.0	0

Example 1.4.1 : *An analog signal is bandlimited to B Hz and sampled at Nyquist rate. The samples are quantized into 4 levels. Each level represents one message. Thus there are 4 messages. The probabilities of occurrence of these 4 levels (messages) are* $p_1 = p_4 = \frac{1}{8}$ *and* $p_2 = p_3 = \frac{3}{8}$. *Find out information rate of the source.*

Solution : (i) To calculate entropy (H) :

We have four messages with probabilities $p_1 = p_4 = \frac{1}{8}$ and $p_2 = p_3 = \frac{3}{8}$. Average information H (or entropy) is given by equation 1.3.5 as,

$$H = p_1 \log_2\left(\frac{1}{p_1}\right) + p_2 \log_2\left(\frac{1}{p_2}\right) + p_3 \log_2\left(\frac{1}{p_3}\right) + p_4 \log_2\left(\frac{1}{p_4}\right)$$

$$= \frac{1}{8}\log_2 8 + \frac{3}{8}\log_2\left(\frac{8}{3}\right) + \frac{3}{8}\log_2\left(\frac{8}{3}\right) + \frac{1}{8}\log_2 (8)$$

$$H = 1.8 \text{ bits / message} \quad \ldots (1.4.2)$$

(ii) To calculate message rate (r) :

We know that the signal is sampled at Nyquist rate. Nyquist rate for B Hz bandlimited signal is,

$$\text{Nyquist rate} = 2B \text{ samples / sec.}$$

Since every sample generates one message signal,

Messages per second,

$$r = 2B \text{ messages / sec.}$$

(iii) To calculate information rate (R) :

Information rate is given by equation 1.4.1 as,

$$R = rH$$

Putting values of r and H in above equation,

$$R = 2B \text{ messages/sec.} \times 1.8 \text{ bits/message}$$

$$= 3.6\, B \text{ bit /sec.} \quad \ldots (1.4.3)$$

Comment :

In the example we discussed above, there are four levels. Those four levels can be coded using binary PCM as shown in Table 1.4.1.

Message or level	Probability	Binary digits
Q_1	$\frac{1}{8}$	0 0
Q_2	$\frac{3}{8}$	0 1
Q_3	$\frac{3}{8}$	1 0
Q_4	$\frac{1}{8}$	1 1

Table 1.4.1

Thus two binary digits (binits) are required to send each message. We know that messages are sent at the rate of $2B$ messages/sec. Hence transmission rate of binary digits will be,

$$\text{Binary digits (binits) rate} = 2 \text{ binits/message} \times 2B \text{ messages/sec.}$$

$$= 4B \text{ binits/sec.}$$

Example 1.4.4 : *A black and white TV picture consists of 525 lines of picture information. Assuming each line consists 525 picture elements, each element having 256 brightness levels and the pictures are repeated at the rate of 30 per sec. Calculate the average information conveyed by a TV set to a viewer.*

Solution : Given data :

One picture contains 525 lines.

One line contains 525 picture elements.

Pictures are sent at 30 per sec.

To obtain picture element rate (r)

$$\text{Picture element rate (r)} = 525 \text{ picture elements/line} \times 525 \text{ lines/picture} \times 30 \text{ pictures/sec}$$

$$= 8.26875 \times 10^6 \text{ picture elements/sec}$$

To obtain entropy of picture elements (H)

Here the picture element can take any one of the 256 brightness levels. All the brightness levels are equally probable. For 'M' number of equally likely messages the entropy of the source is,

$$H = \log_2 M$$

Here M = 256 brightness levels. Hence entropy of the source is,

$$H = \log_2 256 = 8 \text{ bits}$$

To obtain average information (R)

$$\text{Average information rate (R)} = rH = 8.26875 \times 10^6 \times 8$$

$$= 66.15 \text{ Mbps}$$

Extension of Discrete Memoryless Source

Consider the discrete memoryless source having alphabet $X=\{x_1, x_2, ...x_M\}$. This source emits 'M' number of individual symbols. The symbols of this alphabet can occur in groups of two or more. These groups of symbols can be said to have produced from extended source. This extended source is said to have $\{X^n\}$ alphabet with M^n distinct symbols. Here 'n' is the number of successive symbols in one group or block. The symbols generated by the discrete memoryless source are statistically independent. Hence the probability of the group of symbol is obtained by product of individual symbol probabilities. Hence the entropy of X^n and $H(X)$ are related as,

$$H(X^n) = nH(X) \qquad \text{... (1.5.1)}$$

Example : *A zero memory source contains* $X = \{x_1, x_2, x_3, x_4\}$ *with* $P(X) = \left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right\}$

i) Determine entropy of the source.

ii) Determine second extension of the source. Show that $H(X^2) = 2H(X)$.

Solution : (i) To obtain entropy H(X)

The given source alphabet has four symbols and their probabilities are as follows:

$$p(x_1) = p_1 = \frac{1}{2}$$

$$p(x_2) = p_2 = \frac{1}{4}$$

$$p(x_3) = p_3 = \frac{1}{8}$$

$$p(x_4) = p_4 = \frac{1}{8}$$

The entropy is given by equation,

$$H(X) = \sum_{k=1}^{M} p_k \log_2 \frac{1}{p_k} = \sum_{k=1}^{4} p_k \log_2 \frac{1}{p_k}$$

$$= p_1 \log_2 \frac{1}{p_1} + p_2 \log_2 \frac{1}{p_2} + p_3 \log_2 \frac{1}{p_3} + p_4 \log_2 \frac{1}{p_4}$$

Putting the values in above equation,

$$H(X) = \frac{1}{2}\log_2 2 + \frac{1}{4}\log_2 4 + \frac{1}{8}\log_2 8 + \frac{1}{8}\log_2 8$$

$$= \frac{7}{4} \text{ bits/symbol}$$

ii) To obtain second order extension of the source

The source alphabet X contains four symbols. Hence its second order extension will contain sixteen symbols. These symbols, their probabilities and entropy calculations are shown in Table 1.5.1.

i	Second order extension symbol σ_i	Probability of symbol σ_i $p(\sigma_i)$	$p(\sigma_i)\log_2\frac{1}{p(\sigma_i)}$
1.	$x_1 x_1$	$\frac{1}{2}\times\frac{1}{2}=\frac{1}{4}$	$\frac{1}{4}\log_2 4=\frac{1}{2}$
2.	$x_1 x_2$	$\frac{1}{2}\times\frac{1}{4}=\frac{1}{8}$	$\frac{1}{8}\log_2 8=\frac{3}{8}$
3.	$x_1 x_3$	$\frac{1}{2}\times\frac{1}{8}=\frac{1}{16}$	$\frac{1}{16}\log_2 16=\frac{1}{4}$
4.	$x_1 x_4$	$\frac{1}{2}\times\frac{1}{8}=\frac{1}{16}$	$\frac{1}{16}\log_2 16=\frac{1}{4}$
5.	$x_2 x_1$	$\frac{1}{4}\times\frac{1}{2}=\frac{1}{8}$	$\frac{1}{8}\log_2 8=\frac{3}{8}$
6.	$x_2 x_2$	$\frac{1}{4}\times\frac{1}{4}=\frac{1}{16}$	$\frac{1}{16}\log_2 16=\frac{1}{4}$
7.	$x_2 x_3$	$\frac{1}{4}\times\frac{1}{8}=\frac{1}{32}$	$\frac{1}{32}\log_2 32=\frac{5}{32}$
8.	$x_2 x_4$	$\frac{1}{4}\times\frac{1}{8}=\frac{1}{32}$	$\frac{1}{32}\log_2 32=\frac{5}{32}$
9.	$x_3 x_1$	$\frac{1}{8}\times\frac{1}{2}=\frac{1}{16}$	$\frac{1}{16}\log_2 16=\frac{1}{4}$
10.	$x_3 x_2$	$\frac{1}{8}\times\frac{1}{4}=\frac{1}{32}$	$\frac{1}{32}\log_2 32=\frac{5}{32}$
11.	$x_3 x_3$	$\frac{1}{8}\times\frac{1}{8}=\frac{1}{64}$	$\frac{1}{64}\log_2 64=\frac{6}{64}$
12.	$x_3 x_4$	$\frac{1}{8}\times\frac{1}{8}=\frac{1}{64}$	$\frac{1}{64}\log_2 64=\frac{6}{64}$
13.	$x_4 x_1$	$\frac{1}{8}\times\frac{1}{2}=\frac{1}{16}$	$\frac{1}{16}\log_2 16=\frac{1}{4}$
14.	$x_4 x_2$	$\frac{1}{8}\times\frac{1}{4}=\frac{1}{32}$	$\frac{1}{32}\log_2 32=\frac{5}{32}$
15.	$x_4 x_3$	$\frac{1}{8}\times\frac{1}{8}=\frac{1}{64}$	$\frac{1}{64}\log_2 64=\frac{6}{64}$
16.	$x_4 x_4$	$\frac{1}{8}\times\frac{1}{8}=\frac{1}{64}$	$\frac{1}{64}\log_2 64=\frac{6}{64}$

Table 1.5.1 Calculations of second order extension of source

The entropy of the second order extension of the source can be obtained as,

$$H(X^2) = \sum_{i=1}^{16} p(\sigma_i) \log_2 \frac{1}{p(\sigma_i)}$$

Putting value from Table 1.5.1 in above equation,

$$H(X^2) = \frac{1}{2}+\frac{3}{8}+\frac{1}{4}+\frac{1}{4}+\frac{3}{8}+\frac{1}{4}+\frac{5}{32}+\frac{5}{32}+\frac{1}{4}$$
$$+\frac{5}{32}+\frac{6}{64}+\frac{6}{64}+\frac{1}{4}+\frac{5}{32}+\frac{6}{64}+\frac{6}{64}$$
$$= \frac{7}{2} \text{ bits/symbol (extended)}$$

Here observe that,

$$H(X) = \frac{7}{4} \text{ bits/symbol}$$

$$H(X)^2 = \frac{7}{2} \text{ bits/symbol (extended)}$$

The above results confirm the following :

$$H(X)^2 = 2H(X) = 2 \times \frac{7}{4} = \frac{7}{2}$$

Example 1.5.3 : *For DMS 'X' with two symbols x_1 and x_2 and $p(x_1) = 0.9$ and $p(x_2) = 0.1$. Find out the second order extension for the source. Find the efficiency 'η' and redundancy of this extended code.*

Solution : (i) To obtain entropy (H) :

Entropy is given as,

$$H = \sum_{k=1}^{M} p_k \log_2 \frac{1}{p_k}$$

Here $p_1 = 0.9$ and $p_2 = 0.1$. Hence,

$$H = 0.9 \log_2 \left(\frac{1}{0.9}\right) + 0.1 \log_2 \frac{1}{0.1}$$
$$= 0.1368 + 0.3322$$
$$= 0.469 \text{ bits/message.}$$

(ii) To obtain second order extension :

The source alphabet 'X' contains two symbols. Hence its second order extension will contain four symbols. These symbols, their probabilities and entropy calculations are given in Table 1.5.2.

i	Second order extension symbol σ_i	Probability of symbol σ_i $p(\sigma_i)$	$p(\sigma_i) \log_2 \frac{1}{p(\sigma_i)}$
1	$x_1 x_1$	0.9×0.9 = 0.81	$0.81 \log_2 \frac{1}{0.81} = 0.2462$
2	$x_1 x_2$	0.9×0.1 = 0.09	$0.09 \log_2 \frac{1}{0.09} = 0.3126$
3	$x_2 x_1$	0.1×0.9 = 0.09	$0.09 \log_2 \frac{1}{0.09} = 0.3126$
4	$x_2 x_2$	0.1×0.1 = 0.01	$0.01 \log_2 \frac{1}{0.01} = 0.0664$

Table 1.5.2

Entropy of the second order extension of the source can be obtained as,

$$H(X^2) = \sum_{i=1}^{4} p(\sigma_i) \log_2 \frac{1}{p(\sigma_i)}$$

Putting values from above table,

$$H(X^2) = 0.2462 + 0.3126 + 0.3126 + 0.0664$$

$$= 0.9378 \text{ bits/message}$$

Here note that $H(X^2) = 0.938 = 2H(X)$

$$= 2 \times 0.469$$

(iii) To obtain efficiency :

Here no source coding is employed. Here are four symbols in extension of the source. Hence two bits will be required to code four symbols. Hence average number of bits per symbol will be two. i.e., $\overline{N} = 2$ bits/symbol.

$$\text{Code efficiency } \eta = \frac{H(X^2)}{\overline{N}}$$

$$= \frac{0.938}{2}$$

$$= 0.469$$